

Provider Journey Master Documentation

Generated proof-of-concept output. Review before use or publication.

Contents

- Connected Services - Getting Started
 - Connected Services overview
 - Purpose and value
 - Connected services building blocks
 - Authoritative Data Register (ADR)
 - Eligibility Factor Verification (EFV)
 - Secure Data Exchange (SDX)
 - Privacy and security
 - Governance and policy
 - Roles and responsibilities
 - Limits and constraints
 - Support
- APS Infrastructure Platform - SDX How-to Guides
 - Onboarding an Organization
 - Install an Edge Runtime Group
 - Setup Organization Signing
 - Managing Subsystems
 - Managing Services
 - Connecting a Service
 - Connection Resources
 - Event Management
 - Policy Management
- Eligibility Factor Verification TechDoc
 - Introduction

Connected Services - Getting Started

Connected Services overview

Connected Services provides reusable building blocks that help government programs securely share and reuse trusted data.

If you are building or managing a digital service, explore available APIs and datasets below, or learn more about Connected Services and its building blocks.

Find APIs and datasets

Use DevHub search or explore the catalogues to discover:

- APIs that use *Secure Data Exchange* (SDX)
- *Authoritative Data* sources

- Available *Eligibility Factor Verification* (EFVs)

Availability

Connected Services capabilities are introduced iteratively. New APIs, datasets, and components will be added over time.

Learn more

To understand the broader purpose behind these capabilities, review Purpose and value.

To understand how Connected Services components fit together, review Connected Services building blocks.

Purpose and value

Government programs often need information that already exists in another ministry or system.

Today, this can result in delays, duplicate integrations, repeated data collection, inconsistent definitions, and manual verification processes.

Connected Services aims to improve service delivery by reducing duplication, supporting secure reuse of trusted data across government, and strengthening interoperability across ministries.

It does this by establishing common building blocks that enable:

- Access to authoritative data
- Reusable eligibility checks
- Secure, policy-aligned data exchange
- Centralized discovery of APIs and datasets

As these capabilities become available, teams will be able to use (and reuse) standardized components instead of building one-off integrations.

What this means for you

For developers, Connected Services supports:

- Faster integrations with reusable components
- Reduced custom validation logic
- Access to governed and trusted data sources
- Consistent integration guidance and sandbox environments

For product owners and program leads, Connected Services supports:

- Clear data ownership and accountability
- Reduced duplication of datasets
- More consistent eligibility decisions
- Scalable cross-ministry service design

For data custodians, data managers, and data providers, Connected Services supports:

- Clear designation of authoritative data
- Defined access and reuse policies

- Reduced ad hoc data-sharing requests
- Improved visibility into how data is used
- Structured governance processes for oversight and accountability

Connected Services establishes common foundations that support more consistent, secure, and interoperable digital services across government.

Capabilities are being introduced iteratively. Availability of specific components will expand over time.

Explore the components

Connected Services is delivered through reusable building blocks.

To see how these components work together, review Connected Services building blocks.

Connected services building blocks

Connected Services is made up of reusable building blocks that work together to support secure, consistent data reuse across government.

Each building block has a specific role. Together, they reduce duplicate integrations, improve data trust, and support more seamless service delivery.

Authoritative Data Register (ADR)

Authoritative Data is the official source of truth for specific information, such as residency, enrollment, property ownership, or program status.

It provides trusted, governed data that other programs can rely on.

Learn more about ADR.

Eligibility Factor Verification (EFV)

Eligibility Factor Verification (EFV) checks whether the details an applicant provides meet the rules for a particular government program.

For example:

- Is this person a BC resident? Yes/no
- Is this individual enrolled in a specific program? Yes/no

EFVs reduce manual verification and help standardize eligibility checks across programs.

Learn more about EFV.

Secure Data Exchange (SDX)

Secure Data Exchange (SDX) provides a governed way to move approved data between systems.

It ensures that data sharing is secure, policy-aligned, and auditable, without requiring custom integrations for each connection.

Learn more about SDX.

Data and API catalogue

The Catalogue provides a centralized place to discover datasets, APIs, and related documentation.

It supports consistent definitions, improved discoverability, and reuse across ministries.

Explore the Catalogue.

Authoritative Data Register (ADR)

What it is

Authoritative Data is a dataset formally designated as the trusted source for a specific purpose within government.

The ADR service provides the governance process that identifies, assesses, certifies, and oversees these datasets.

Authoritative status is contextual. A dataset is authoritative for **a defined scope and use case**, not for all purposes.

Who this is for

ADR supports both those who publish authoritative data and those who integrate with it.

Primary audience

- Developers building integrations
- Solution architects designing cross-ministry services

Secondary audience

- Data custodians
- Product owners
- Program and policy leads
- Data managers

When to use authoritative data

Use Authoritative Data when:

- You need a trusted, government-approved dataset
- You are building eligibility, verification, or automation logic
- You are integrating across ministries or systems
- You want to reduce duplication or avoid shadow datasets
- You need clarity on definitions, ownership, and accountability

Do not assume that all datasets are authoritative. Always confirm status before integrating.

How ADR works (high level)

ADR formalizes how datasets are reviewed and certified.

1. A data custodian submits a dataset for review.
2. The dataset is assessed against authoritative data criteria.

3. A governance body reviews and endorses (or declines) certification.
4. Approved datasets are listed in the Authoritative Data Directory.
5. Certified datasets are monitored and maintained over time.

Authoritative certification is not a one-time approval. It requires ongoing oversight.

Where to go next

TODO Add Links when available

- Authoritative Data Certification Process
- Authoritative Data Directory
- Data Register Standards
- Glossary and Data Dictionary

Eligibility Factor Verification (EFV)

What it is

Eligibility Factor Verification (EFV) is the administrative process of confirming that the information provided by an applicant is accurate and meets the rules for a specific government benefit or service.

It standardizes how programs verify deterministic facts such as income thresholds, residency status, or identity.

EFV serves as the bridge between authoritative data and a program's eligibility rules. It helps confirm deterministic facts like income thresholds, residency status, and identity.

The EFV framework supports:

- Connection to trusted and authoritative data sources
- Reduction of manual interpretation and document review
- Faster, more consistent decision support
- A defensible evidence trail for explainability and audit

Who this is for

EFV supports both those building eligibility systems and those responsible for program outcomes.

Primary audience

- Developers building eligibility tools and integrations
- Solution architects designing automated adjudication systems

Secondary audience

- Product owners managing benefit programs
- Program and policy leads defining eligibility criteria
- Adjudicators and intake staff using decision-support tools

When to use Eligibility Factor Verification

Use EFV when:

- You need to confirm a specific, verifiable fact (e.g., “Is this person a B.C. resident?”)
- You want to replace manual document collection and reviews with trusted automated data signals
- You require a defensible audit trail explaining how a determination was reached

EFV is best suited for deterministic checks.

It should not be used for complex cases requiring significant human discretion or contextual judgment unless those patterns are clearly defined and governed.

How it works (high level)

1. A program identifies the specific eligibility factors required (e.g., income, residency).
2. The system connects to an Authoritative Data source.
3. A verification check compares applicant information to the trusted dataset.
4. The system returns a verification signal (e.g., Pass/Fail) along with metadata such as source and verification timestamp.
5. Adjudicators use the signal to support automated approval or focus on exceptions requiring review.

Verification is most effective when it reduces the need for applicants to upload scans or photographs of physical documents.

Where to go next

TODO Add Links when available

- Eligibility Factor Verification Standards
- Authoritative Data Directory
- Trust Metadata and Lineage Guidelines

Secure Data Exchange (SDX)

What it is

Secure Data Exchange (SDX) serves as the secure exchange layer within Connected Services, enabling the controlled and protected transfer of data between participating organizations.

The exchange operates behind the scenes to ensure that when a citizen’s information is sent or received, the data remains protected throughout the process - from initial exchange to storage.

SDX is built around two foundational components:

- **Security Controls**

This includes encryption, authentication, and data integrity mechanisms that protect information in transit and verify participating systems.

- **Governance and Compliance**

This includes auditing and logging functionality, policy enforcement, non-repudiation, and timestamping to support oversight and accountability.

Together, these components ensure that sensitive data exchanged through SDX maintains confidential, protected, and compliant with B.C.'s security and regulatory requirements.

Who this is for

SDX is intended for any organization participating in Connected Services that needs to securely exchange sensitive data across organizational boundaries.

This includes organizations such as LTSA and ICBC, as well as other ministries and partners.

Participating organizations may act as data providers, data consumers, or both.

When to use Secure Data Exchange

SDX is required when sensitive data (e.g., personal, financial, or health-related information) is being shared across organizational boundaries.

Use SDX when:

- Data exchange requires enhanced security controls due to sensitivity of the data
- Policy-based access enforcement is needed
- Auditability and non-repudiation are required

SDX is not required for public or non-sensitive data that does not require cross-boundary security controls.

How it works (high-level)

SDX operates through an onboarding and registration process within the SDX ecosystem.

At a high level:

1. The SDX Operator registers their organization.
2. An Organization Administrator assigns the necessary System Owner role(s).
3. The System Owner(s) register(s) services and subsystems and is able to browse the service catalogue.
4. An OpenAPI Specification (OAS) file is submitted for API registration. This spec must be in YAML format.
5. Once onboarded, an Edge Server is then provisioned.
6. Secure connections are established between participating systems within the SDX ecosystem.

All exchanges are authenticated, authorized, and logged automatically.

Detailed onboarding steps and technical configuration guidance are available in SDX Onboarding – TechDocs. [TODO Unstable link!](#)

Relationship to privacy and security

SDX provides the technical implementation of the privacy and security controls described in Privacy and Security.

Review Privacy and security for information on governance alignment and accountability structures.

Where to go next

- SDX Onboarding – TechDocs TODO Unstable link!

Privacy and security

Connected Services is built on a secure, policy-driven trust foundation.

Data remains with the ministry that owns it. *Connected Services* does not centralize or store program data. Instead, it enables secure, auditable exchange between trusted systems.

Privacy and security controls are embedded into the technical design, not added afterward.

Secure exchange by design

All data exchange within *Connected Services* occurs through *Secure Data Exchange* (SDX), which enforces:

- Authentication of participating systems
- Verification that requests are unaltered
- Real-time access control
- Timestamped logging of transactions

These controls ensure that only authorized, policy-compliant requests are fulfilled.

For technical details on how SDX implements these controls, including request verification and Edge Server architecture, see *Secure Data Exchange* (SDX).

Data ownership and control

Connected Services does not take ownership of ministry data.

- Ministries remain custodians of their data
- Access decisions are enforced based on predefined policies
- Data is exchanged directly between approved participants
- No centralized data repository is created

This model supports interoperability while preserving program accountability.

Identity and access enforcement

Access is controlled at request time using short-lived, scoped tokens.

Each request must:

- Be authenticated
- Meet defined access policies
- Be validated before any data is exchanged

Access enforcement is automated and consistent across participants.

Logging, audit, and transparency

All transactions are logged automatically.

- Requests are timestamped

- Actions are auditable
- Records support compliance and oversight
- Logging enables accountability without exposing underlying data

This design supports transparency while minimizing data exposure.

Scaling securely

As Connected Services expands, the same trust and security foundation applies.

- New partners onboard under shared trust rules
- Policies are enforced consistently
- Security controls do not need to be rebuilt for each integration

This supports scalable, secure interoperability across ministries.

How decisions are defined

Privacy and security controls enforce rules — but those rules are established through governance and policy direction.

To understand how oversight, accountability, and decision-making structures shape Connected Services, review Governance and policy.

Governance and policy

Connected Services relies on clear governance and policy direction to support responsible, consistent data sharing across ministries.

Governance defines how data is designated, accessed, exchanged, and overseen. It establishes who is accountable, how decisions are made, and how policies are enforced across Connected Services components.

Some governance structures are already in place. Others will continue to evolve as Connected Services expands and additional capabilities are introduced.

Transparency about how decisions are made and how data is used is a core principle of Connected Services.

Evolving governance model

Connected Services is being introduced through an initial use case, with governance and policy frameworks actively being established to support that implementation.

Governance structures, standards, and decision processes are being defined as part of the foundational work required to launch the first connected service.

As additional capabilities are introduced over time, governance processes will continue to mature and expand to support broader adoption.

Governance decisions, standards, and supporting documentation will be published as they are formalized.

What governance covers

Governance in Connected Services applies across:

- *Authoritative Data* designation and certification
- Access control and authorization rules
- Shared standards for interoperability and data exchange
- Logging, oversight, and accountability mechanisms
- Policy alignment across participating ministries

These governance areas ensure that data reuse occurs in a consistent, auditable, and policy-aligned manner.

Roles and accountability

Governance does not remove data ownership from ministries.

- Data custodians remain responsible for their data
- Access decisions are governed by defined policies
- Oversight bodies review and endorse authoritative designations
- Technical controls enforce policy decisions consistently

Clear accountability across technical and program areas is essential to maintaining trust.

Relationship to privacy and security

Governance and policy defines the rules and accountability structures.

Privacy and security describes how those rules are technically enforced through *Secure Data Exchange* (SDX) and related controls.

Review Privacy and security for information on technical enforcement mechanisms.

Who participates

Governance depends on clearly defined roles across ministries and technical teams.

To see how different actors contribute to publishing, accessing, and overseeing data, review Roles and responsibilities.

Roles and responsibilities

Connected Services involves multiple roles across technical, program, and governance areas.

Clear roles and responsibilities are essential to ensure that data is published, accessed, maintained, and exchanged responsibly.

As Connected Services is implemented through its initial use case, role definitions and associated responsibilities will continue to be refined and published. Shape

Data providers (Suppliers)

Data providers are responsible for managing and publishing authoritative data.

Roles may include:

- Data Custodian
- Data Manager
- Data Architect
- Developer
- Data Publisher / Editor

Responsibilities typically include:

- Maintaining data quality and accuracy
- Defining appropriate use and access conditions
- Participating in authoritative data designation (where applicable)
- Ensuring compliance with governance and policy requirements

Data consumers (Subscribers)

Data consumers use authoritative data or reusable components within their applications and services.

Roles may include:

- Developer
- Solution Architect
- Application Manager
- Product Owner
- Policy Lead
- Data Analyst
- Data Scientist
- Program Staff

Responsibilities typically include:

- Integrating using approved and governed mechanisms
- Adhering to defined access policies
- Using data only for approved purposes
- Ensuring appropriate handling of sensitive information

Shared Responsibilities

Connected Services requires collaboration between providers and consumers.

Shared responsibilities include:

- Aligning on definitions and standards
- Respecting data ownership and accountability
- Participating in governance processes
- Supporting audit and transparency requirements

Evolving role clarity

As governance processes mature and additional building blocks are implemented, role definitions and associated responsibilities will be further defined and documented.

Updates will be published as structures are formalized.

Working within current scope

Roles and responsibilities continue to evolve alongside Connected Services implementation.

To understand the current boundaries and phased rollout of capabilities, review Limits and constraints.

Limits and constraints

Connected Services is being introduced through an initial use case and will expand iteratively over time.

While the long-term direction includes broader cross-ministry integration and life-event-based service delivery, current capabilities are focused on establishing foundational building blocks.

Current scope

At this stage:

- Implementation is focused on a defined MVP use case
- Not all ministries or datasets are onboarded
- Not all processes are consolidated
- Some discovery and integration workflows may remain fragmented
- Governance and standards are actively being refined

Connected Services provides improved structure and consistency compared to previous approaches, but full consolidation and standardization will occur over time.

Incremental expansion

As additional building blocks and use cases are implemented:

- More datasets and APIs will become available
- Discovery and catalogue functionality will mature
- Governance processes will be formalized and published
- Cross-ministry participation will expand

Capabilities will grow iteratively rather than through a single comprehensive launch.

What this means for teams

Teams should:

- Expect phased availability of components
- Confirm authoritative status and access rules before integration
- Monitor updates to documentation and governance standards
- Engage early when planning integrations that depend on emerging capabilities

Support

Support is available for teams working with *Connected Services*.

Connected Services brings together multiple components and supporting teams, so not all questions will go to the same place. Use the guidance below to help route your request.

Not sure where to go? Start with one of the common scenarios below.

Common starting points:

- API or integration issues: APS support
- Understanding Connected Services: Connected Services documentation
- Specific API or dataset: Record page guidance

When to seek support

Seek support if:

- you are unsure whether a dataset or API is appropriate for your use case
- you need help understanding how to access or use an API
- you are blocked in onboarding, setup, or integration
- you need clarification on technical documentation
- you are not sure which team owns a component or process
- you need help finding the correct next step in DevHub

API platform and integration support

For support related to API platform services, onboarding, access, gateway setup, or technical issues, use the APS support guidance:

API Platform Services support

This is the primary source for current APS support channels and instructions.

Documentation and discovery support

If you are using Connected Services documentation to understand what components exist, how they fit together, or where to begin, you can:

- review the relevant Connected Services pages in this TechDoc
- use DevHub search to locate related APIs, datasets, and technical documentation

If you are still unsure where to go, start with the APS support page above so your request can be routed appropriately.

Questions about a specific API or dataset

If you are reviewing a specific API or dataset, first check the record page for:

- ownership information
- related documentation
- access guidance
- linked contacts or support paths

Where record-level support details exist, use those first.

Troubleshooting and when you're blocked

If you're unable to proceed or something isn't working as expected:

- confirm you are using the correct API or dataset
- review documentation and access requirements
- check for available contacts or support guidance on the record page
- use APS support if you are blocked on platform, onboarding, or integration steps

Before you ask for support

When possible, include:

- the API, dataset, or component name
- a link to the relevant DevHub page
- a short description of what you are trying to do
- where you are blocked
- any error message or unexpected behaviour
- the environment, if relevant

Providing this information helps support teams direct requests and respond more efficiently.

Related information

You may also want to review:

- Connected Services overview
- Connected Services building blocks
- Privacy and security
- Governance and policy
- Roles and responsibilities

APS Infrastructure Platform - SDX How-to Guides

Onboarding an Organization

This page shows how to onboard an organization onto the Secure Data Exchange.

The steps described in this page are performed by the following Organization roles:

Role	Function
SDX Operator	Establish member organizations and assign legal representatives Org Admin role
Organization Admin	Manage System Admin role assignment for the organization
System Admin	Manage subsystem onboarding for the particular organization

Use cases:

- Register new organization

- System Admin role assignment
- List organizations
- Assign gateway for organization

Register new organization

This is performed by the SDX Operator to onboard a new Organization.

Restish CLI

Help information about the organization creation/update operation:

```
restish aps put-organization
```

Example:

```
echo '
{
  "name": "my-org",
  "title": "My Org",
  "description": "It is an organization for me",
  "extSource": "custom",
  "extRecordHash": "0000",
  "tags": [
    "member_class:MIN", "member_id:MYORG"
  ],
  "orgUnits": []
}
' | restish aps put-organization ca.bc.gov
```

Reference

- **API PUT** /organizations/{org}

Parameters: {org}=ca.bc.gov

```
{
  "name": "ministry-of-food",
  "title": "Ministry of Food",
  "description": "It is a ministry concerned with food",
  "extSource": "custom",
  "extRecordHash": "0000",
  "tags": [
```

```

    "member_class:MIN",
    "member_id:FOOD"
  ],
  "publicBodyId": null,
  "orgUnits": []
}

```

tags is array<string>. An array containing an object is rejected by the live request schema.

System Admin role assignment

This is performed by the Organization Admin to assign system administrators access to manage their systems.

The organization-scoped role that grants this access is system-admin, which carries permission System.Manage for the organization. This is distinct from any system-level roles and from organization-admin, which carries GroupAccess.Manage, Namespace.Assign, and Dataset.Manage instead.

Restish CLI

Help information about the organization role membership synchronization operation:

```
restish aps put-organization-access
```

Example:

```

echo '
{
  "name": "my-org",
  "parent": "/ca.bc.gov",
  "members": [
    {
      "member": {
        "email": "aidan.cope@gov.bc.ca"
      },
      "roles": ["system-admin"]
    }
  ]
}
' | restish aps put-organization-access ca.bc.gov

```

Reference

- API PUT /organizations/{org}/access

Parameters: {org}=ministry-of-food

```
{
  "name": "ministry-of-food",
  "parent": "/ca.bc.gov",
  "members": [
    {
      "member": {
        "email": "janis@testmail.com"
      },
      "roles": ["system-admin"]
    }
  ]
}
```

List organizations

Retrieve the list of organizations available in the SDX catalog.

Restish CLI

```
restish sdx organization-list
```

Reference

- API GET /catalog/organizations

```
json title="Response Body" [ { "name": "ministry-of-food", "title":
"Ministry of Food", "description": "It is a ministry concerned with food",
"member": { "memberClass": "MIN", "memberId": "FOOD" } } ]
```

Get organization details

Retrieve the details of an organization available in the SDX catalog, optionally including the organization's RBAC role membership.

Restish CLI

Help information about the operation:

```
restish sdx organization-get
```

Example call:

```
restish sdx organization-get my-org
```

Include role membership:

```
restish sdx organization-get my-org --include-access
```

Note: includeAccess does not require authentication

organization-get has always been callable without a bearer token, and includeAccess does not change that: role membership is resolved using the platform's own Keycloak service credentials, not the caller's - so passing includeAccess=true returns the organization's role members to anonymous callers, the same way the base listing already returns every organization's name/title/member details to anonymous callers.

Assign gateway for organization

An organization will have public keys that it will use for organization signing of traffic through SDX. Each organization is assigned a unique gateway where the public keys are managed.

Restish CLI

Help information about the operation to assign a runtime group:

```
restish sdx register-organization-gateway
```

Example:

```
restish sdx register-organization-gateway \  
my-org
```

Next steps

- Install an Edge Runtime Group

Install an Edge Runtime Group

This page shows how to install a runtime group for your organization on SDX.

Before your systems can start to connect with other systems in SDX, your organization must either deploy a runtime group in your own infrastructure (client-hosted), or you must signup for using one of the shared runtime groups (community-hosted).

To learn more about the communication protocol between a client Edge Runtime Group and a service Edge Runtime Group, visit the SDX Data Access Protocol document.

The steps described in this page are performed by the following roles:

Role	Function
System Admin	Request a new runtime group, and manage onboarding a new runtime group

Note: Community Hosted

If you are going to use one of the community-hosted runtime groups, please reach out to the APS team, and skip this how-to guide.

Use cases for client-hosted:

- Establish a new runtime group
- Register a runtime group gateway
- Deploy runtime group infrastructure
 - Request a one-time-use certificate signing token
 - Deploy the runtime group infrastructure
 - Apply default routes and controls
 - Verification test
 - Add public key to the registry

Prerequisites

- Install Restish CLI
- Install Helm (if deploying the runtime group infrastructure)

Establish a new runtime group

To establish a runtime group, you need to know the internet-facing IP address that will be used to route traffic to this runtime group.

Restish CLI

Help information about the operation to list available runtimes:

```
restish sdx create-runtime-group
```

Example:

```
restish sdx create-runtime-group \  
my-org \  

```

```
'name: newrg, environment: lab, hostedOrganizations: ["my-org"],
sdxEndpoint: "https://142.34.194.118:443"'
```

Reference

This is performed by a System Admin to create a new runtime group.

- **API** PUT /organizations/{org}/runtime-groups

Parameters:

- {org}=<your-organization>

```
{
  "name": "abc123",
  "environment": "dev",
  "sdxEndpoint": "https://142.34.194.118:443",
  "consumerEndpoint": "http://internal.abc123.servers.sdx",
  "hostedOrganizations": ["ministry-X", "ministry-Y"]
}
```

Attribute	Description
name	Unique identifier (lowercase alphanumeric text between 3 and 8 characters)
environment	Target environment
sdxEndpoint	Routable IP-based endpoint from the internet (example above is the Gold ingress IP)
consumerEndpoint	Domain that the Runtime Group uses automatically (port 8000, internal.)
hostedOrganization	List of all the organizations that are permitted to use this particular Runtime Group

Register a runtime group gateway

As a System Admin, you perform this task. Once complete, you can set up the default routing policies for this runtime group.

Warning: Registration is not a safe retry

register-runtime-group-gateway is create-only. If the runtime group's namespace was already partially registered (for example after an earlier failed or interrupted attempt), rerunning registration for the same name fails rather than repairing the existing namespace, and manually deleting the namespace can leave retained Kong catalog services/routes with no corresponding live data-plane configuration. There is currently no documented preflight or reconcile command for this state — verify with the APS team before deleting an existing runtime namespace to retry registration.

Restish CLI

Help information about the operation to assign a runtime group:

```
restish sdx register-runtime-group-gateway
```

Example:

```
restish sdx register-runtime-group-gateway \  
my-org newrg
```

Reference

- **API PUT** /organizations/{org}/runtime-groups/{name}/gateway

Parameters:

- {org}=<your-organization>
- {name}=<your-runtime-group-name>

An assigned Gateway ID will be returned. This Gateway can be used to configure default routes and controls for this runtime group.

Note: Granting namespace access to additional users

Registration grants SDX namespace scopes only to the caller who created the runtime group's gateway. There is no Restish or other APS/SDX REST operation for granting or repairing another user's scopes on an existing namespace. Additional users must be granted access through the API Services Portal's **Administration Access** page (GraphQL API), which itself requires the requesting user to hold Namespace.Manage on that namespace. Do not attempt to grant access by creating an ordinary Keycloak authorization permission directly — the platform expects a resource-owner-managed UMA permission ticket, and the

two are not interchangeable. A namespace with no `Namespace.Manage` holder currently has no self-service recovery path; contact the APS team.

Deploy runtime group infrastructure

Request a one-time-use certificate signing token

The runtime group infrastructure uses a token from the CA to bootstrap the first certificate.

The certificate is used for supporting mTLS between runtime groups.

This is performed by a System Admin to request a new cert signing token.

Restish CLI

Help information about the operation:

```
restish sdx generate-one-time-use-token
```

Example call:

```
restish sdx generate-one-time-use-token \
  my-org newrg lab

# Generate and save to "token"
restish sdx generate-one-time-use-token \
  myo newrg lab | jq -r .token > token
```

Reference

- **API** POST `/organizations/{org}/runtime-groups/{name}/environments/{environment}/tokens`

Parameters:

- `{org}`=<your-organization>
- `{name}`=<your-runtime-group-name>
- `{environment}`=<target-environment>

It will return a token which can be extracted and stored in a local file for the next step.

Deploy the runtime group infrastructure

We have a helm chart available for deploying a runtime group into a Kubernetes/Openshift environment.

There has been some exploratory work for deploying infrastructure in Azure.

Please reach out to the APS team to discuss your requirements if the helm chart is not sufficient.

```
export IP="<ip specified in the sdxEndpoint above>"
export EDGE_ID="<name specified above>"
export ENV=lab
export DOMAIN="${EDGE_ID}.${ENV}.servers.sdx"

helm upgrade --install ${EDGE_ID} \
  --set bootstrap.tls.token=$(cat token) \
  --set bootstrap.tls.cn=${DOMAIN} \
  --set bootstrap.tls.ip=${IP} \
  --set route.host=${DOMAIN} \
  oci://ghcr.io/bcgov/aps-devops/sdx-edge:0.3.7

# If you want to upgrade to a newer helm chart version, you can run
helm upgrade --install ${EDGE_ID} \
  --reset-then-reuse-values \
  --set bootstrap.tls.token="" \
  oci://ghcr.io/bcgov/aps-devops/sdx-edge:0.3.7
```

Provision default routes and controls

You can now call the API to preview and then publish Gateway configuration containing the default routing rules for the runtime group.

Actions available:

- `preview` : see what configuration the pattern produced
- `apply` : apply the configuration
- `diff` : dry run showing what will be updated if the `apply` is used
- `delete` : deletes the configuration

Restish CLI

Help information about the operation to generate and apply Gateway configuration:

```
restish sdx provision-config-from-pattern
```

Example:

```
restish sdx provision-config-from-pattern \
  my-org sdx-runtime-group.r1 \
  --action apply \
  'parameters:{ runtimeGroupName: newrg, environment: lab }'
```

Note: Required authorization scope

The generated help for `provision-config-from-pattern` lists `System.Manage` as the required scope. The effective requirement is actually `GatewayPattern.Publish`, resolved against the runtime's namespace (for example `<namespace>:GatewayPattern.Publish`), and it is granted automatically to the runtime's registering System Admin. Authorization happens in two stages: `GatewayPattern.Publish` gates the `preview/diff/apply/delete` endpoint itself for the human caller, and a separate `GatewayConfig.Publish` scope, held by the internal `sdx-provisioner` client rather than the caller's own token, gates the downstream GWA namespace publish that `diff` and `apply` trigger. `preview` returns generated configuration before that downstream request and therefore only needs the outer `GatewayPattern.Publish` scope.

Note: Reading diff results

`diff` is a dry run, but its response reuses mutation-sounding fields — a top-level `applied` count and a per-provider `status: applied` — even when nothing was changed. Those fields describe successful **processing** of the dry run, not the number of gateway changes committed. Do not treat a `diff` response as evidence that Kong configuration changed; check the nested `details.message` (for example `Dry-run. No changes applied.`) and the `Created/Updated/Deleted` summary for the actual proposed changes, and use `apply` to commit them.

Verification test

Running the following should return 400 No required SSL certificate was sent.

```
curl -v -k --resolve ${DOMAIN}:443:${IP} \
https://${DOMAIN}
```

You can verify the consumer internal endpoint by opening a terminal on the runtime group Kong pod and running:

```
curl -v --resolve internal.${DOMAIN}:8000:127.0.0.1 \
http://internal.${DOMAIN}:8000/hello
```

Note: Peer TLS trust

These checks verify the runtime group's own edge, but do not verify trust between peer runtime groups. Before relying on an active peer-to-peer connection, confirm that the calling edge's Kong trusts the peer edge's issuing CA (Kong returns HTTP 502 with an upstream TLS verification failure otherwise). Also note that the endpoint host displayed by the portal may be normalized by GWA to the namespace's permitted environment domain rather than shown verbatim.

Add public key to the registry

The public key will be used for other runtime groups to verify the integrity of the request.

The helm deployment and bootstrap job will create the sdx-edge secret for the tls certificate pair. Save the `tls.crt` contents to a `tls.crt` file locally.

Restish CLI

Help information about the operation:

```
restish sdx provision-config-from-pattern
```

Example call:

```
restish sdx provision-config-from-pattern \
  my-org sdx-keys.r1 \
  --action apply \
  'parameters:{ certificatePem[0]: @tls.crt, runtimeGroupName: newrg,
environment: lab }'
```

Reference

Using the same pattern endpoint from above, you can use the `sdx-keys.r1` pattern to add the public key using the certificate from the runtime group.

```
{
  "pattern": "sdx-keys.r1",
  "parameters": {
    "runtimeGroupName": "<runtime-group-name>",
    "environment": "lab|dev|test|prod",
    "certificatePem": ["<public-certificate-pem-format>"]
  }
}
```

`certificatePem` is an array; only the public certificate is supplied here, and the private key must remain mounted in the runtime group's edge. `organization` is derived from the `{org}` path parameter and does not need to be supplied in the body.

Warning: Prerequisite for signed connections

Registering the runtime group's public key with `sdx-keys.r1` is a mandatory prerequisite before enabling the `sign` upgrade on a consumer connection or the `verify` upgrade on a provider connection (see [Connecting a Service](#)). The `trust-sign` plugin embeds the

runtime's JWKS URI in the signature but does not create or publish the key set itself — until `sdx-keys.r1` has been applied, the JWKS URL will return 404 Key set not found and traffic relying on verification will fail.

Runtime Group management

Decommission Runtime Group

To be documented..

Steps to decommission:

- uninstall infrastructure
- remove default routes
- remove keys
- delete runtime group

Next steps

- Setup Organization Signing

Setup Organization Signing

Organization signing is used to cryptographically verify that messages were transmitted with permission between the involved systems using an authorized Runtime Group.

Note: Setup is Optional

There are various upgrades related to the connection that can be enabled. Performing the counter-sign using the organization keys is one of these upgrades. See Connection Resources for more information.

If policy requires this to be enabled, then follow the steps to setup organization signing. The organization must register signing keys with the Runtime Group that their systems are using to connect to SDX.

Role	Function
Organization Admin	Manage organization keys for signing

Use cases:

- Request a new signing key CSR
- Get the CSR signed by an approved Certificate Authority
- Add CA signed certificate to registry

Prerequisites

- Install Restish CLI

Request a new signing key CSR

Restish CLI

Help information about the operation:

```
restish sdx create-new-key
```

Example call:

```
restish sdx create-new-key \  
my-org \  
environment: lab, runtimeGroupName: newrg
```

The inputs for the CSR will be derived from your organization details and the runtime group you are registering it on.

You will get back a document in YAML format similar to this:

```
signing_algorithm: ECDSA_SHA_512  
csr: |  
-----BEGIN CERTIFICATE REQUEST-----  
MIIBWTCB/wIBADBGMQswCQYDVQQGEwJDQTFCEAGA1UECgw5TWluaXN0cnkgb2Yg  
Q2l0aXplbnMgU2VydmVjZXMvc2VyaWFsTnVtYmVyPUxBQj9NSU4vU0hBUKUwMQ0w  
CwYDVQQDDARDsVRaMFkwEwYHKoZIzj0CAQYIKoZIzj0DAQcDQgAEyqHYa+yG6YGE  
y1fR1gUEaRzhbe3REt10BC6F9JDstvROUuYBaKYZJbXZ6wQ8q+bwDRzlcGv1Bc/k  
a73T0xd7XqA9MDsGCSqGSIb3DQEJJDjEuMCwwKgYDVR0RBCMwIYIfbGFilWlpbilj  
aXR6LnNoYXJlMC5zZXJ2ZXJzLnNkeDAKBggqhkJOPQQDAgNJADBGAiEA2VFX1pKP  
0FYl+JNux0Xz+E1CLEcNk9Acy3pJH4e/cmACIQDi0q/xxg598GYBQ0c+gQtiCsPL  
ubWazfkHoChChFcX1g==  
-----END CERTIFICATE REQUEST-----  
jwk: |  
'{"y": "TlLmAWimGSW12esEPKvm8A0c5XBr9QXP5Gu909MXe14", "kid": "RYDLAYlWr184FTwRk21jQzvZS03U0woqRKN5lH  
yG6YGEy1fR1gUEaRzhbe3REt10BC6F9JDstvQ", "kty": "EC"}'  
pub_key: |  
-----BEGIN PUBLIC KEY-----  
MFkwEwYHKoZIzj0CAQYIKoZIzj0DAQcDQgAEyqHYa+yG6YGEy1fR1gUEaRzhbe3R  
Et10BC6F9JDstvROUuYBaKYZJbXZ6wQ8q+bwDRzlcGv1Bc/ka73T0xd7Xg==  
-----END PUBLIC KEY-----  
inputs:  
  org_name: Ministry of Citizens Services  
  requester_name: unknown  
  serial_number: LAB/MIN/SHARE0  
  requester_email: unknown  
  san: lab-min-citz.share0.servers.sdx
```

```
country: CA
common_name: CITZ
```

Get the CSR signed by an approved Certificate Authority

SDX Operator provides a Certificate Authority (CA).

Reach out to the APS team with your CSR to get it reviewed, approved and signed.

Add CA signed certificate to registry

Once you receive back the certificate, save the certificate (and its intermediate CAs) in a new .crt file, and the root certificate for the CA in root.crt.

Use one of the root certificates from below depending on your environment.

For details about environments, visit [SDX Environments](#).

Playground

```
-----BEGIN CERTIFICATE-----
MIIBqjCCAVCgAwIBAgIRA0x+wUXYgyRY9VGzArB7V+swCgYIKoZIzj0EAwIwMzEx
MC8GA1UEAxMoQ1NCQyBTZW1cmUgRGF0YSBFeGNoYW5nZSBBUFNNUU1QgUm9vdCBD
QTAEFw0yNjA3MTYxNTE4MzBaFw0zNjA3MTMxNTE4MzBaMDMxMTAvBgNVBAMTKENT
QkMgU2VjdXJlIERhdGEgRXhjaGFuZ2UgQVBTVFNUIFJvb3QgQ0EwWTATBgqhkhj0
PQIBBgqhkhj0PQMBBwNCAAQceMa6kWDVEqVnG9wZwD7zn3y67LGICe8lBiQTzxRR
h5LdqhryGnKIOsUMMujuvFUTsuUsaG9KywMh/NuTMkaLMo0UwQzA0BgNVHQ8BAf8E
BAMCAQYwEgYDVROTAQH/BAGwBgEB/wIBAAdBgNVHQ4EFgQUUnb0hKEg01Xkv1zeZ
zCXusaEzQHEwCgYIKoZIzj0EAwIDSAARQIhAPue7wdiw/MocolfSBGZgEp4V03Q
5Ws2v0C/BBRWI+UTAIaQnQJxusfbu3t0VXjTNg4Z/+uH+bGIQgIMPasQamFQSQ==
-----END CERTIFICATE-----
```

Staging

```
-----BEGIN CERTIFICATE-----
MIIBozCCAUqgAwIBAgIRA0qrFwxuBQzATeE2ybv4ci8wCgYIKoZIzj0EAwIwMDEu
MCwGA1UEAxMlQ1NCQyBTZW1cmUgRGF0YSBFeGNoYW5nZSBERVYgUm9vdCBDQTAE
Fw0yNjA3MTYxNTE4MzBaFw0zNjA3MTMxNTE4MzBaMDMxMTAvBgNVBAMTJUNTQkMg
U2VjdXJlIERhdGEgRXhjaGFuZ2UgREVWIFJvb3QgQ0EwWTATBgqhkhj0PQIBBgqh
khj0PQMBBwNCAASKZrREActpsjEdst6vKcQmxEe060uVnoBQ7luxWymcSosJJCHD
WEV/2e9EyGPLHpw5RstPgX+Ha5D6+BcKGzjio0UwQzA0BgNVHQ8BAf8EBAMCAQYw
EgYDVROTAQH/BAGwBgEB/wIBAAdBgNVHQ4EFgQUYb2Jz7MuAOKY8bu9NM6tjvS6
xkYwCgYIKoZIzj0EAwIDRWAwRAIgTFXSb8bq5Z8P8oIC03BVHkHxCm0GRcQL10TL
GtlSuWYCIBPfZrhbZX4oFhEk0sq7HXlBJuh6Zaa6dcs03RIUt1Gm
-----END CERTIFICATE-----
```

Non-Prod

```
-----BEGIN CERTIFICATE-----
MIIBPtCCAUugAwIBAgIQBYjllY/VB9YWcN4ix0J9/jAKBgqhkhj0PQQDAjAxMS8w
```

```
LQYDVQQDEyZDU0JDIFNlY3VyZSBYXZBIEV4Y2hhbmdlIFRFRU1QgUm9vdCBDQTAe
Fw0yNjA3MTcyMTM5MTBaFw0zNjA3MTQyMTM5MTBaMDExLzAtBgNVBAMTJKNTQkMg
U2VjdXJlIERhdGEgRXhjaGFuZ2UgVEVTVCBSb290IENBMFkwEwYHKOZIZj0CAQYI
KoZIZj0DAQcDQgAEDGVAeDHUeLVBGxuZP7PS/Z8c1RZBb4z8+S7qRySm8VQ06Qt9
X9VTps3N8rXSk1L015ELWT0WbWBAf0xi0om5MaNFMEMwDgYDVR0PAQH/BAQDAgEG
MBIGA1UdEwEB/wQIMAYBAf8CAQIwHQYDVR00BBYEFayGhiqDdh67pQepVugBJ35P
zupYMAoGCCqGSM49BAMCA0gAMEUCIQC0Z98oClh1Ngi63m9Jwvib4GUcfP7b894o
MGzh5gHWSAIgHZw/p+dnqJiq/ukUIUWrWxTuNmuvGIYG+/HscgaV+yM=
-----END CERTIFICATE-----
```

Prod

```
-----BEGIN CERTIFICATE-----
MIIBPjCCAUyGAWIBAgIRAMJMz93KjTyWx4EVe4SLeAEwCgYIKoZIzj0EAwIwMTEv
MC0GA1UEAxMmQ1NCQyBTZW51cmUgRGF0YSBFeGNoYW5nZSBQUk9EIEFJvb3QgQ0Ew
HhcNMjYwNzIxMjIzMzAzWhcNMzYwNzE4MjIzMzAzWjAxMS8wLQYDVQQDEyZDU0JD
IFNlY3VyZSBYXZBIEV4Y2hhbmdlIFB00QgUm9vdCBDQTBZMBMGByqGSM49AgEG
CCqGSM49AwEHA0IABNj3iloPdVoyfNAFAodzv7Fcz4Pm8l2fYxeP9hZdA8wpk2J
n0mK9hlqburDu3ehXREbEnZmDClxqhwc/bfSxSjRTBDMA4GA1UdDwEB/wQEAwIB
BjASBgNVHRMBAf8ECDAGAQH/AgECMB0GA1UdDgQWBbTntqkvHMI2ZafMGoKv+NQy
jIx4XDAKBggqhkJOPQDAgNIADBFAiAFg+IddYG6zgqalUXVxT+PDfBgI/Jns2Mc
jS6ocTR5cwIhAKL/0ak7GhWLPsbKxR7wo0z+Qe7CzPsgidM8+SFH5d
-----END CERTIFICATE-----
```

You will then be able to use this information to update the public key details with the new certificate in the JWKS registry.

Restish CLI

Help information about the operation:

```
restish sdx provision-config-from-pattern
```

Example call:

```
restish sdx provision-config-from-pattern \
  my-org sdx-keys.r1 \
  --action apply \
  'parameters:{ environment: lab, certificatePem[0]: @new.crt, caCerts:
  @root.crt }'
```

Managing Subsystems

Overview

This page shows how to manage subsystems on the Secure Data Exchange.

The steps described in this page are performed by the following roles:

Role	Function
System Admin	Manage systems and service catalog entries for the particular organization

Use cases:

- Register a subsystem
- Assign your subsystem to a runtime group
- Subsystem management
 - Administer subsystem RBAC
 - Privacy zones and Common SSO
 - Delete a subsystem

Prerequisites

- Install Restish CLI

Register a subsystem

Restish CLI

Help information about the operation:

```
restish sdx upsert-subsystem
```

Example call:

```
restish sdx upsert-subsystem \  
my-org \  
name: MY-NEW-SUBSYSTEM
```

Reference

API PUT /organizations/{org}/subsystems

Parameters: {org}=ministry-of-food

```
{  
  "name": "MY-NEW-SUBSYSTEM"  
}
```

Assign your subsystem to a runtime group

As a System Admin, you perform this task. Once complete, you can set up routing policies for connecting to other systems on SDX.

Restish CLI

Help information about the operation to list available runtimes:

```
restish sdx list-runtime-groups
```

Example:

```
restish sdx list-runtime-groups \  
my-org \  
--filter available
```

Help information about the operation to assign a runtime group:

```
restish sdx register-subsystem-gateway
```

Example:

```
restish sdx register-subsystem-gateway \  
my-org MY-NEW-SUBSYSTEM \  
runtimeGroupName: newrg
```

An assigned Gateway ID will be returned. This Gateway can be used to configure routes and controls for services it connects to.

Reference

To find available runtime groups for your organization, use the following API:

- **API** GET /organizations/{org}/runtime-groups?filter=available

Parameters:

- {org}=<your-organization>

After choosing a runtime group, make a note of the name.

If there are none returned, reach out to the SDX Operator (APS Team) to find out information for onboarding your organization onto SDX.

You can now call the API to assign your subsystem to the runtime group.

- **API PUT** /organizations/{org}/subsystems/{name}/gateway

Parameters:

- {org}=<your-organization>
- {name}=<subsystem-name>

```
json title="Request Body" {  "runtimeGroupName": "<runtime-group-name>" }
```

An assigned Gateway ID will be returned. This Gateway can be used to configure routes and controls for services it connects to.

Warning: Confirm the runtime currently hosts your organization

Registration currently succeeds with HTTP 200 even if the chosen runtime group's hostedOrganizations no longer includes your organization — for example after a runtime's hosting was changed after you queried list-runtime-groups. The resulting gateway is created but is missing the runtime's host domain, which will cause confusing downstream failures when publishing services or provider patterns. After registering, call get-subsystem-client and confirm the runtime group actually appears in the returned runtimeGroups before proceeding.

Warning: Registration is create-only

register-subsystem-gateway cannot be used to repair or reconcile an already-registered subsystem namespace — repeating it against an existing namespace returns HTTP 422 Namespace already exists, even after the underlying problem (such as the missing hosted-organization relation above) has been corrected. Deleting the gateway to retry is not a safe workaround for a deterministic SDX gateway ID: it removes the resource set and marks the Keycloak namespace group decommissioned while the subsystem continues to reference the same gateway ID, and the namespace name will still be rejected as existing on a retry. Contact the APS team for recovery rather than deleting and recreating.

Subsystem management

Administer subsystem RBAC

A subsystem's creator is granted all three roles automatically when assigning the subsystem to a runtime group. Use this operation afterward to add, change, or remove role membership - for example when a colleague joins the team or the creator leaves the organization.

The supported roles are:

Role Name	Role ID	Functions
Subsystem Owner	subsystem-owner	Overall accountable owner for the subsystem
Tech Lead	tech-lead	Technical point of contact for the subsystem
Access Manager	access-manager	Manages which clients have access to services

Warning: Updating access replaces the full member list

put-subsystem-access is a full sync, not an incremental grant: any member/role combination not included in the request body is **revoked**. To add one person without affecting anyone else, first call get-subsystem-access and include its existing members in your update alongside the new one.

Restish CLI

Help information about the operation to get current access:

```
restish sdx get-subsystem-access
```

Example:

```
restish sdx get-subsystem-access \  
my-org SUBSYSTEM-NAME
```

Help information about the operation to update access:

```
restish sdx put-subsystem-access
```

Example - grants Janis all three roles and Mark access-manager only (and revokes any other role membership not listed here):

```
echo '
{
  "members": [
    {
      "member": { "email": "janis@testmail.com" },
      "roles": ["subsystem-owner", "tech-lead", "access-manager"]
    },
    {
      "member": { "email": "mark@gmail.com" },
      "roles": ["access-manager"]
    }
  ]
}
' | restish sdx put-subsystem-access my-org SUBSYSTEM-NAME
```

Confirm the change:

```
restish sdx get-subsystem-access \
my-org SUBSYSTEM-NAME
```

Submitting a role name other than subsystem-owner, tech-lead, or access-manager returns a 4xx naming the unsupported value.

Service Provider Privacy Zone

If the subsystem is going to be a Resource Server (RS) providing a service, the subsystem **MUST** set its privacy zone defined in the Authorization Party (AP) and be reviewed by the AP Owner before it can be connected to clients.

For a non-exhaustive list, see privacy zones.

Restish CLI

```
restish sdx upsert-subsystem \
my-org \
name: MY-NEW-SUBSYSTEM, \
privacyZone: "urn:ca:bc:gov:buseco:prod"
```

Service Client Integration

If the subsystem is going to be a Relying Party, the subsystem **MUST** set the Authorization Party (AP) Integration ID to identify which access requests this subsystem will accept from the AP.

Restish CLI

Example setting the integration ID to 22308.

```
restish sdx upsert-subsystem \  
  my-org \  
  name: MY-NEW-SUBSYSTEM, \  
  'integrations: [{integrationClientId:"22308"}]'
```

Delete a subsystem

A subsystem can be deleted when it has no active connection requests and no gateway configuration.

If deletion succeeds, related OAS services are deleted with the subsystem.

Restish CLI

Help information about the operation:

```
restish sdx delete-subsystem
```

Example:

```
restish sdx delete-subsystem \  
  my-org SUBSYSTEM-NAME
```

Reference

API DELETE /organizations/{org}/subsystems/{name}

Parameters:

- {org}=<your-organization>
- {name}=<subsystem-name>

The delete request will not proceed if any of the following are true:

- the subsystem has active connection requests as a client
- a service under the subsystem has active connection requests
- subsystem gateway configuration exists

After a subsystem is deleted, the same subsystem name can be used again.

Next steps

- Managing Services

Managing Services

Overview

This page shows how to manage services on the Secure Data Exchange.

The steps described in this page are performed by users with the following roles:

Role	Function
System Admin	Organization-level role for managing subsystems and services
Subsystem Owner	Subsystem-level role for managing services for a subsystem

Use cases:

- Register a service
- View API service catalog
- Subsystem management
 - Delete a service

Prerequisites

- Install Restish CLI

Register a service

To register a service, you need to identify the environment you are deploying the service to, the upstream URL for routing to where your service is running, and the OpenAPI specification itself.

For details about valid environment values, visit [Environment labels](#).

Restish CLI

Help information about the operation:

```
restish sdx upsert-oas-service
```

Example:

```
restish sdx upsert-oas-service \  
  my-org \  
  --subsystem MY-NEW-SUBSYSTEM \  
  --environment lab \  
  --upstream-url "https://your-upstream-endpoint" \  
  --openapi-spec https://your-openapi-spec
```

```
--rsh-header "Content-Type: application/yaml" \  
< openapi.yaml
```

Note: OAS security scopes are not automatically enforced

Distinct OpenAPI security scopes declared per-operation (read, create, update, delete, etc.) are **not** automatically converted into route-level runtime authorization. Registering an OAS that declares scopes does not by itself restrict which operations a caller with a valid token can reach. Runtime access control must be configured explicitly through connection upgrades — see the token, consumerMatch, and tokenExchange options in Connection Resources and the JWT Keycloak plugin — and, as deployed today, the JWT guard verifies the token's signature, issuer, expiry, and sub, but does not itself verify scope, audience, or authorized party. Treat declared OAS scopes as descriptive metadata until per-operation enforcement is explicitly configured.

Troubleshooting service registration

An upsert-oas-service request can return HTTP 503, validation_service_unavailable, OAS validation service unavailable for two different reasons that look identical to the caller:

- the validation service is genuinely unreachable or timed out, or
- the validation service is reachable and responded, but its internal rules engine failed to process the document (for example, a Spectral rule throwing on a legitimate null value in the document).

Both cases currently surface as the same 503. If you hit this error:

- retry once to rule out a transient network/timeout issue;
- if it persists, check whether the validator's /versions endpoint is reachable and returning 200 — if it is, the failure is more likely in the rules engine than in service availability;
- if the failure appears tied to a specific document construct (for example, a nullable field with an explicit null example value), report it to the APS team with the offending OAS fragment rather than continuing to retry.

View API service catalog

Restish CLI

List all subsystems:

```
restish sdx subsystems-list
```

List only name and title of APIs:

```
restish sdx list-service-catalog | jq '.[] | .name+": "+.title'
```

Retrieve a service's OpenAPI Description (OAD)

The registered OAD for a service can be retrieved two ways:

- publicly, with no credentials required:

```
GET /ds/api/sdx/v1/catalog/services/{serviceId}/oas-spec
```

- organization-scoped, requiring `System.Manage`, via the `get-organization-service-spec` Restish operation.

Neither operation returns the fully SDX-transformed publication artifact described by the SDX API Standard. The stored document is validation-enriched only — registration adds `x-csbc-api-standard` and `x-csbc-api-standard-ruleset` under `info`, recording the validation service's version and ruleset — but the `servers` section, OAuth endpoints, and `provider.info.contact` are otherwise returned as uploaded, and no SDX `externalDocs` is added. Do not assume the retrieved document's `servers` or OAuth URLs are environment-correct without further transformation.

`x-csbc-api-standard` here records the validation service version, not a provider-declared API standard release name; treat its meaning as implementation-specific until this is reconciled with the SDX API Standard.

An interactive Swagger UI is not currently provided through SDX for a registered service; only the OAD retrieval operations above are available.

Service management

Delete a service

A service can be deleted when there are no active connection requests for it.

Restish CLI

Help information about the operation:

```
restish sdx delete-organization-oas-service
```

Example:

```
restish sdx delete-organization-oas-service \
  my-org SERVICE-NAME
```

The delete request will not proceed if the service has active connection requests.

After a service is deleted, the same service name can be used again.

Next steps

- Connecting a Service

Connecting a Service

This page shows how to make a connection between your system and another on the Secure Data Exchange.

SDX supports different authorization models, which are identified by a `policyVersion` when a connection request is submitted. The following policies are supported:

Policy	Description
SDX.R0.00	Simple peer-to-peer connection policy. Requires a <code>requesterDetails</code> object (see below). Available only in the SDX Playground.
SDX.R1.00	Adds Common SSO, token-exchange and scopes support; requires additional requester and gateway resources and is not a drop-in default.

This guide focuses on the `SDX.R0.00` policy where it establishes a basic peer-to-peer connection. It is only available in the SDX Playground environment.

Policy `SDX.R1.00` is part of a collaboration with Common SSO (CSS) where CSS integrations will be able to request API and scope access through their tool and have the connection request managed in SDX for approvals. SDX will send events to CSS when connections have been approved/revoked so that the appropriate scopes can be provisioned in Keycloak. For clients that call services across privacy zones, SDX will perform an exchange of the token so that the subject identifier properly identifies the user in the target privacy zone.

The steps described in this page are performed by the following roles:

Role	Function
Tech Lead	For service clients, request/revoke connections to another service
Access Manager	For service providers, review and approve client connection requests

Use cases:

- Request access (as consumer)
- Review connection access requests
- Approve access (as provider)
- Open a connection
 - Consumer side
 - Provider side
- Connection management

- Delete a connection request

Prerequisites

- Install Restish CLI

Request access (as consumer)

Restish CLI

Help information about the operation:

```
restish sdx upsert-connection
```

Example call:

```
restish sdx upsert-connection \  
  my-org \  
  clientId: MIN.MYORG.MY-NEW-SUBSYSTEM, \  
  serviceId: LAB.MIN.MYORG.EFV-ICBC.v0, \  
  policyVersion: SDX.R0.00, \  
  "requesterDetails: {}"
```

Note: requester details

Under SDX.R0.00, requesterDetails must be supplied together with policyVersion on this initial request — when both are present, the controller replaces requesterDetails.requester with the authenticated caller's name and email. Omitting requesterDetails leaves an invalid empty default in place that will pass creation but fail activation. requesterDetails can only be set on creation; it cannot be repaired afterwards through upsert-connection (that field is reserved for the provisioner on update) — if activation fails for a missing requester record, deactivate and delete the connection, then recreate it with policyVersion and requesterDetails included together.

Review connection access requests

Restish CLI

Help information about the operation:

```
restish sdx list-connections
```

Example call:

```
restish sdx list-connections \  
my-org
```

Approve access (as provider)

Restish CLI

Help information about the operation:

```
restish sdx update-connection-approval
```

Example call:

```
restish sdx update-connection-approval \  
my-org \  
clientId: MIN.MYORG.MY-NEW-SUBSYSTEM, \  
serviceId: LAB.MIN.MYORG.EFV-ICBC.v0, \  
isApproved: true, isActive: true
```

Open a connection

Once the connection request has been approved, both sides are able to publish the routing rules for opening a channel between the two systems.

Consumer side

Restish CLI

Help information about the operation:

```
restish sdx upsert-connection
```

Prepare a pattern input file (client-input.json) for the Consumer:

```
{  
  "gatewayPatterns": {  
    "sdx-p2p-consumer.r1": {  
      "upgrades": {  
        "sign": {},  
        "verify": {}  
      }  
    }  
  }  
}
```

```
}  
}
```

Example call:

```
restish sdx upsert-connection \  
my-org \  
clientId: MIN.MYORG.MY-NEW-SUBSYSTEM, \  
serviceId: LAB.MIN.MYORG.EFV-ICBC.v0, \  
clientResources: @client-input.json
```

For details on configuring the `sdx-p2p-consumer.r1` pattern, go to [Connection Resources](#).

Provider side

Restish CLI

Help information about the operation:

```
restish sdx upsert-connection
```

Prepare a pattern input file (`service-input.json`) for the Provider:

```
{  
  "gatewayPatterns": {  
    "sdx-p2p-provider.r1": {  
      "upgrades": {  
        "mtlsAuth": {},  
        "sign": {},  
        "verify": {}  
      }  
    }  
  }  
}
```

Example call:

```
restish sdx upsert-connection \  
my-org \  
clientId: MIN.MYORG.MY-NEW-SUBSYSTEM, \  
serviceId: LAB.MIN.MYORG.EFV-ICBC.v0, \  
serviceResources: @service-input.json
```

For details on configuring the `sdx-p2p-provider.r1` pattern, go to [Connection Resources](#).

Note: Associating an OAuth integration client with a connection

`requesterDetails.client.clientId` (used above for `sdx-p2p-consumer-access.r1/ACL` and, when configured, `consumerMatch`) is provisioner-managed. It cannot be written directly through the SDX management API — attempting a direct `upsert-connection` update of `requesterDetails` returns HTTP 400. Registering an `integrationClientId` on the consumer subsystem (see [Register an SDX Subsystem](#)) is only the first half of the relationship: the field is populated by calling the supported `POST /v1/integrations/{clientId}/access-requests` operation on the SDX Partner Authorization Services API, which validates the requested scopes against the registered OAD and writes the requester metadata with trusted provisioner credentials. That API uses a separate base URL and bearer-token audience from the main SDX API — consult the APS team for the current Restish/API configuration for it. Before enabling strict `consumerMatch`, verify that the connection contains the integration client and that the Kong consumer created by `sdx-p2p-consumer-access.r1` exists, since enabling `consumerMatch` ahead of a matching consumer blocks legitimate traffic.

Connection management

Deleting a connection

Deleting a connection request is two steps. First step is to make it inactive:

```
restish sdx upsert-connection \  
  my-org \  
  clientId: MIN.MYORG.MY-NEW-SUBSYSTEM, \  
  serviceId: LAB.MIN.MYORG.EFV-ICBC.v0, \  
  isActive: false
```

Then the connection can be deleted using its unique identifier.

Restish CLI

Help information about the operation:

```
restish sdx delete-connection
```

Example call:

```
restish sdx delete-connection \  
  ministry-of-citz 2010
```

Reference

- **API DELETE** /organizations/{org}/connections/{id}

Parameters:

- {org}=<your-organization>
- {id}=<connection-request-id>

The {org} value can be the consumer organization or the provider organization for the connection request.

Successful response:

```
{
  "result": "deleted",
  "id": "1"
}
```

If gateway configuration still exists on either side, the request is rejected. Remove the remaining gateway configuration and try again.

Connection Resources

Connection resources for peer-to-peer will have different rules over time around how they should be configured. This page describes all the parameters that are available.

Note: These patterns are not invoked directly

sdx-p2p-consumer.r1, sdx-p2p-consumer-access.r1, sdx-p2p-provider.r1, and their upgrades are **connection resources**, not gateway patterns invoked through the public /patterns endpoint or provision-config-from-pattern. You configure them by setting `clientResources.gatewayPatterns` (consumer side) and `serviceResources.gatewayPatterns` (provider side) on a connection via `upsert-connection`, as shown in [Connecting a Service](#). The provisioner evaluates them automatically whenever the connection's `isActive` state changes — there is no separate preview/publish/delete step for these patterns, and deleting a peer-to-peer configuration is done by setting `isActive: false` on the connection rather than by deleting the pattern directly.

All Parameters

- The following parameters **MUST** be set: `clientId`, `serviceId`

Parameter	Type	Rule
clientId	string	required
serviceId	string	required
isApproved	boolean	optional; default=false
isActive	boolean	optional; default=false
requesterDetails	object	optional
clientResources	object	optional
.gatewayPatterns	object	optional
serviceResources	object	optional
.gatewayPatterns	object	optional

isApproved is set by a user with the “Access Manager” subsystem role.

requesterDetails

Parameter	Type	Rule
.client	object	optional
.client.integrationId	string	optional
.client.clientId	string	optional
.client.privacyZone	string	optional
.requester	object	required
.requester.name	string	required
.requester.email	string	optional
.scopes	set of string	optional
.service	object	optional
.service.clientId	string	optional
.submissionId	string	optional

clientResources.gatewayPatterns

Parameter	Type	Rule
sdx-p2p-consumer-access.r1	object	required
.integrationClientId	string	optional, example 325
sdx-p2p-consumer.r1	object	required
.tlsVerify	boolean	optional, default true
.stripPath	boolean	optional, default false
.clientRuntimeOverride	string	optional, example MIN.CITZ.pzgw
.upgrades	object	optional

sdx-p2p-consumer upgrades

Parameter	Type	Rule
sign	object	optional
verify	object	optional
counterSign	object	optional
dpop	object	optional
token	object	optional
.allowedAud	string	optional
.allowedIss	string[]	required
.scope	string	optional
.consumerMatch	boolean	optional
.consumerMatchClaim	string	optional
.consumerMatchClaimCustomId	boolean	optional
.consumerMatchIgnoreNotFound	boolean	optional
acl	object	optional
tokenExchange	object	optional
.clientId	string	required
.tokenEndpoint	string	required, any value
.scopes	string[]	optional
.audience	string	optional

serviceResources.gatewayPatterns

Parameter	Type	Rule
sdx-p2p-provider.r1	object	required
.upstreamUrl	string	optional
.upgrades	object	optional

sdx-p2p-provider upgrades

Parameter	Type	Rule
mtlsAuth	object	optional
mtlsAcl	object	optional
sign	object	optional
verify	object	optional
counterSign	object	optional
token	object	optional
.allowedAud	string	optional
.allowedIss	string[]	required
.scope	string	optional
.consumerMatch	boolean	optional
.consumerMatchClaim	string	optional
.consumerMatchClaimCustomId	boolean	optional
.consumerMatchIgnoreNotFound	boolean	optional
acl	object	optional

SDX.R1.00 Policy

The SDX.R1.00 policy adds Common SSO tokens for client authentication, token-exchange for crossing privacy zones, and resource scopes.

Service Client

Both `sdx-p2p-consumer-access.r1` and `sdx-p2p-consumer.r1` patterns are required.

The following upgrades to `sdx-p2p-consumer.r1` are required:

Upgrade	Purpose
token	Verify access token issued by an approved IAM provider (authorized party: client integration)
acl	Verify client represents the SDX client subsystem
tokenExchange	Connected Services Link - Initiate token exchange for cross-privacy zone requests
sign	Standard Edge Runtime token added as an X-Edge-Token header
verify	Verification of Edge Runtime token on response from Provider
counterSign	Client organization transaction signature on request

Service Provider

`sdx-p2p-provider.r1` pattern is required.

The following upgrades to this pattern are required:

Upgrade	Purpose
token	Verify access token issued by an approved IAM provider (authorized party: SDX)
sign	Standard Edge Runtime token added as an X-Edge-Token response header
verify	Verification of Edge Runtime token on request from Client
counterSign	Service organization transaction signature on response

Event Management

This page shows how to publish an AsyncAPI service that can be used to publish messages and for consumers to setup webhooks to receive messages.

Warning: Preview

This feature is in preview only, which means it is experimental. It is available in our LAB environment as is.

The steps described in this page are performed by the following roles:

Role	Function
Subsystem Owner	Subsystem-level role for managing services for a subsystem

Use cases:

- Register an async service
- Configure a publisher endpoint
- Connecting a service

- Configure a webhook
- Publishing a message

Prerequisites

- Install Restish CLI

Register an async service

Restish CLI

Help information about the operation:

```
restish sdx upsert-oas-service
```

Example:

```
restish sdx upsert-oas-service \
  ministry-of-citz \
  --subsystem MY-NEW-SUBSYSTEM \
  --rsh-header "Content-Type: application/json" \
  < asyncapi.yaml
```

Configure a publisher endpoint

This sets up a protected URL that can be used by the Resource Server to publish messages on the topics described in the AsyncAPI.

Restish CLI

Help information about the operation:

```
restish sdx provision-config-from-pattern
```

Example call:

```
restish sdx provision-config-from-pattern \
  ministry-of-citz events-publisher.r1 \
  --action apply \
  parameters:{serviceId:"LAB.USR.ACOPE.HELLO-WORLD-APPLICATION.v0"}
```

The output from this call will be a `publisher_url` that the Resource Server will be able to securely call to send messages.

Connecting a service

With the async service published, clients can request access to the AsyncAPI service.

All the same security controls that are in place for OpenAPIs, are also available for AsyncAPIs.

The how to guide for making connections is at [Connecting a Service](#).

Configure a webhook

Once a connection has been approved, the client is able to configure the webhook details so that it can start to receive messages from the publisher.

Restish CLI

Help information about the operation:

```
restish sdx provision-config-from-pattern
```

Example call:

```
echo '{
  "parameters": {
    "connId": "42",
    "clientId": "LAB.MIN.CITZ.SDG-FE",
    "serviceId": "LAB.USR.ACOPE.HELLO-WORLD-APPLICATION.v0",
    "webhookUrl": "https://bright-island-08.webhook.cool"
  }
}' | \
restish sdx provision-config-from-pattern \
  ministry-of-citz \
  --action apply
```

Publishing a message

The RS published an AsyncAPI spec to state all the events it will publish.

The RS will use its local runtime group publisher endpoint to send a message.

The client authentication will use the same approach described in the [SDX Data Access Protocol](#) document.

```
curl -v -H "Host:internal.share0.servers.sdx" \
  http://localhost:8000/sdx/1/815a243837865c1ed61e94c0/messages \
  -H "Content-Type: application/json" \
  -d '{"value":{"note":"return it!"}}'
```

The return, if successful, is a 202 Accepted, payload:

```
{
  "topicName": "Event-LAB.USR.ACOPE.HELLO-WORLD-APPLICATION.v0",
  "partition": 0,
  "errorCode": 0,
  "baseOffset": "0",
  "logAppendTime": "-1",
  "logStartOffset": "0"
}
```

Policy Management

This page shows how to create OPA and CEDAR policies and use them for enforcement on SDX.

Warning: Preview

This feature is in preview only, which means it is experimental. It is available in our LAB environment as is.

The steps described in this page are performed by the following roles:

Role	Function
System Admin	Organization-level role for managing subsystems and services
Subsystem Owner	Subsystem-level role for managing services for a subsystem

Use cases:

- Register a policy
- Upgrade connection with a PEP
- Register a data source

Register a policy

NOTE: The package name in the policy should match the `subsystem_id` (with `.` and `-` changed to `_`) For example: package `LAB_USR_ACOPE_APS_KAFKA.authz`

```
{
  "pattern": "opal-policy.r1",
  "parameters": {
    "subsystem_id": "LAB.USR.ACOPE.APS-KAFKA",
    "name": "authz",
    "policy": {{POLICY}}
  }
}
```

Inputs

When designing your policies, assume the pep plugin will provide the following data about the particular request:

```
{
  "method": "GET",
  "path": "/v1/widgets/1234",
  "named_params": {
    "id": "1234"
  },
  "token": {
    "sub": "111",
    "aud": "abc",
    "scopes": "scope1 scope2"
  }
}
```

Upgrade connection with a PEP

When configuring the `sdx-p2p-*.r1` gateway patterns for consumer and provider, use the pep upgrade to run the particular policy that you created in the previous step.

```
{
  "upgrades": {
    "pep": {
      "policy_name": "authz"
    }
  }
}
```

Register a data source

The data source (policy information point) is a powerful way of pulling in authorization data to be used in the decision making. The `upstream_url` should be accessible from the runtime group that the subsystem is registered on.

```
{
  "pattern": "opal-data-source.r1",
  "parameters": {
    "subsystem_id": "LAB.USR.ACOPE.APS-KAFKA",
    "name": "user-gateways",
    "upstream_url": "https://httpbun.com/any"
  }
}
```

For referencing the data source in your policy, use the `subsystem_id` package convention:

```
import data.tenant["LAB.USR.ACOPE.APS-KAFKA"] as dat

items := {row |
  row := dat["user-gateways"]}
}
```

Eligibility Factor Verification TechDoc

Introduction

Eligibility Factor Verification, or EFV, helps government programs check specific facts used in eligibility decisions.

Many programs need to confirm similar things, such as income, residency, or other information an applicant provides. Today, this can involve manual document review, screenshots, forms, or staff follow-up. EFV is intended to reduce that effort by helping programs check certain facts against approved or trusted data sources, where available.

EFV does not decide if someone is eligible for a program. EFV is not a program rule engine. It does not approve or deny applications. It helps check facts that a program may use in its own decision process. The program still owns its eligibility rules and makes the final decision.

EFV can support a program by returning a verification result and related information, such as the source used, when the check happened, and any limits or caveats that apply. Program staff or systems can then use that result to support a decision, focus review on exceptions, or reduce the need for applicants to provide documents.

EFV is in early development. The first focus is income verification. Future eligibility factors may be explored over time based on program need, source availability, agreements, policy authority, privacy and security requirements, and technical feasibility.

Example (in a call outbox) =====

A program may need to confirm whether an applicant meets an income rule.

Without EFV, the applicant may need to upload tax documents, and staff may need to review them manually.

With EFV, the program may be able to request an income verification check through a trusted source. The program receives a result and supporting details. The program then applies its own rules and decides what happens next.

Getting Started

Use this page to decide whether EFV may be useful for your program or service.

EFV is still developing. The best first step is to contact the EFV team with a clear use case. The team can help determine whether EFV is a good fit, whether an existing verification factor is available, and what other teams or approvals may be involved.

When EFV may be useful

EFV may be useful when your program needs to check a specific fact used in an eligibility decision.

EFV may help when your program wants to:

- Use a trusted source instead of applicant-submitted documents
- Reduce manual document review or staff follow-up
- Support more consistent decisions
- Keep a record of what was checked, when, and against what source
- Help staff focus on exceptions or higher-risk cases
- Reuse a verification pattern that may also help other programs

Understanding data-use requirements

Using EFV may require more than a technical connection. Before information can be requested or used, the consuming program and data provider may need to confirm the applicable legal authority, privacy requirements, agreements, consent or notice requirements, security controls, and business rules.

These requirements will depend on the program, the eligibility factor, the data source, and the intended use. The EFV team can help identify the requirements and teams that may need to be involved. The consuming program remains responsible for confirming that it has the authority and approvals needed to request and use the information.

What makes a good EFV candidate

A good EFV candidate is usually a verification need that is specific, repeatable, and connected to a trusted source.

EFV may be a stronger fit when:

- The same fact is checked often
- The current process is manual
- The information is not already easy to access or reuse
- The verification pattern could help more than one program

When EFV may not be the right fit

EFV may not be the right fit when:

- The decision depends on case-by-case judgment
- The fact cannot be checked through a trusted source
- The program does not have authority to request or use the data
- The information is already available through an existing dataset, API, service, or public source
- The program needs broad data access instead of a specific verification need
- The program needs policy design support, rather than verification support

Before you contact the EFV team

You do not need a full solution before contacting the EFV team.

Start with the eligibility fact you need to check, how you check it today, and what problem you are trying to solve.

If you already know the potential data source or provider, intended use, legal authority, consent or notice requirements, privacy and security considerations, or agreement needs, include those as well. You do not need to have every requirement resolved before contacting the EFV team.

Eligibility factors

Use this page to understand how EFV looks at possible eligibility factors.

An eligibility factor is a specific fact that a program may need to check before making a decision. This could include income or other information used in program rules.

EFV is organized around these factors. Each factor may use a different source, need different approvals, or have different limits. Because of this, EFV does not treat every factor the same way.

The first EFV factor is income verification. EFV is starting with income verification as the first reusable verification capability that programs may be able to consume.

Future factors may be explored over time based on program need, source availability, agreements, policy authority, privacy and security requirements, and technical feasibility.

Not every eligibility fact needs EFV. Some facts may already be supported through an existing dataset, API, service, or program process. EFV is intended for verification needs that are specific, repeatable, and not already easy for programs to access or reuse.

Before a factor can be supported through EFV, the EFV team needs to understand:

- What fact the program needs to check
- Which program rule or workflow the check supports
- Whether there is a trusted source for the information
- Whether the program has authority to request or use the information
- What legal authority, agreements, consent or notice requirements, privacy protections, security controls, audit requirements, or other provider conditions apply

- Whether the data is current enough for the program's use case
- Whether the verification pattern could be reused by other programs
- Whether there is a practical technical path to access or verify the information

Data providers need confidence that their information will only be requested and used for an approved purpose and under the required conditions. As part of developing an eligibility factor, the EFV team can help identify what requirements may need to be supported. This may include technical, legal, privacy, consent, agreement, and business requirements. This may include determining whether a required authorization or consent check needs to occur before a verification request is sent. The specific approach will depend on the provider, program, data source, and approved use case.

A factor should not be presented as available until the source, permitted purpose, authority, agreement needs, technical path, and support needs are understood well enough to guide users.

Income Verification

Income verification is the first eligibility factor in development through EFV.

This page provides factor-specific information for programs that may need to check income as part of an eligibility process.

Status

Income verification is **in development** as the first EFV use case.

What income verification may support

Income verification may support programs that need to check income-related information as part of an eligibility decision or review process.

The type of response EFV returns will depend on the approved use case and source requirements.

For example, a program may need to confirm whether an applicant meets an income rule without asking the applicant to upload tax documents for manual review.

What the response does not mean

An income verification response does not decide whether someone is eligible for a program.

The program remains responsible for:

- Defining eligibility rules
- Applying program policy
- Deciding how the response is used

- Managing exceptions
- Completing any required staff review
- Making the final eligibility decision

Program readiness

Before using income verification, a program may need to confirm:

- The income rule or program requirement being supported
- The purpose for using the income information
- The authority to request or use the information
- Consent requirements
- Privacy and security requirements
- Agreement or MOU requirements
- Audit and monitoring expectations
- How long records or consent evidence must be retained
- Who can access the result
- How exceptions, denied responses, or incomplete matches will be handled

Depending on the approved use case and source requirements, the income verification process may also need to confirm that required authorization, consent, or other conditions have been met before a request is sent. The EFV team will work with the consuming program, data provider, and appropriate privacy, legal, security, and technical teams to identify the required approach.

Caveats and limitations

Income verification has source-specific requirements and constraints.

Programs should not assume that income verification can be used until the authority, agreement path, consent model, permitted data elements, technical approach, and operational support expectations are confirmed.

The EFV team can help determine whether income verification may be a fit for a program use case and what discovery is needed next.

Support and Next Steps

EFV is still developing. Support channels and service expectations will be added as they are confirmed.

To discuss whether EFV may be a fit for your program or service, contact the EFV team. (Email? Justin? Teams Channel?)

Include the program name, the eligibility factor you want to verify, the current process, and any timeline or pilot dependency.

Related resources

Links will be added as they become available.